

1. Implement a java program to demonstrate creating an ArrayList, adding elements, removing elements, sorting elements of ArrayList. Also illustrate the use of toArray() method.

```
import java.util.ArrayList;
import java.util.Collections;

public class ArrayListDemo {
    public static void main(String[] args) {

        //Creating an ArrayList
        ArrayList<String> list = new ArrayList<>();

        //Adding elements
        list.add("Banana");
        list.add("Apple");
        list.add("Mango");
        list.add("Orange");

        System.out.println("Original List:");
        System.out.println(list);

        //Removing an element
        list.remove("Mango"); // remove by value
        System.out.println("\nAfter Removing 'Mango':");
        System.out.println(list);

        //Sorting elements
        Collections.sort(list);
        System.out.println("\nAfter Sorting:");
        System.out.println(list);

        //Using toArray() method
        Object[] arr = list.toArray();

        System.out.println("\nArray elements using toArray():");
        for (Object obj : arr) {
            System.out.println(obj);
        }
    }
}
```

OUTPUT

Original List:

[Banana, Apple, Mango, Orange]

After Removing 'Mango':

[Banana, Apple, Orange]

After Sorting:

[Apple, Banana, Orange]

Array elements using toArray():

Apple

Banana

Orange

2. Develop a program to read random numbers between a given range that are multiples of 2 and 5, sort the numbers according to tens place using comparator.

```
import java.util.ArrayList;
import java.util.Collections;
import java.util.Comparator;
import java.util.Random;

public class TensPlaceSortExample {

    public static void main(String[] args) {

        //Given range
        int start = 10;
        int end = 100;

        Random r = new Random();
        ArrayList<Integer> list = new ArrayList<>();

        //Generate 15 random numbers
        for (int i = 0; i < 15; i++) {

            int num = r.nextInt(end - start + 1) + start;

            //Check if number is multiple of 2 and 5
            if (num % 2 == 0 && num % 5 == 0) {
                list.add(num);
            }
        }

        System.out.println("Multiples of 2 and 5:");
        System.out.println(list);
    }
}
```

```
// Sort based on tens place using Comparator
Collections.sort(list, new Comparator<Integer>() {
    public int compare(Integer a, Integer b) {

        int tensA = (a / 10) % 10; // get tens digit
        int tensB = (b / 10) % 10;

        return tensA - tensB; // ascending order
    }
});

System.out.println("After Sorting by Tens Place:");
System.out.println(list);
}
```

OUTPUT

Multiples of 2 and 5:

[40, 70, 20, 90, 30]

After Sorting by Tens Place:

[20, 30, 40, 70, 90]